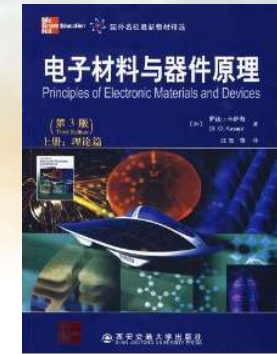
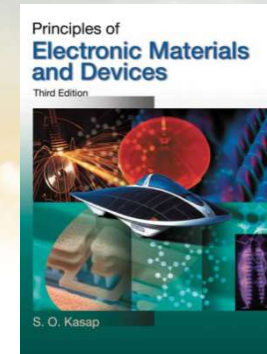




南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《固态电子学》 Solid-State Electronics



Ch3-3 Elementary Quantum Physics

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Some slides are modified from other Profs' PPTs. Their contributions are greatly acknowledged.

Free Electron

$$\frac{d^2 \varphi(x)}{dx^2} + \frac{2m}{\hbar^2} (E - V) \varphi(x) = 0$$

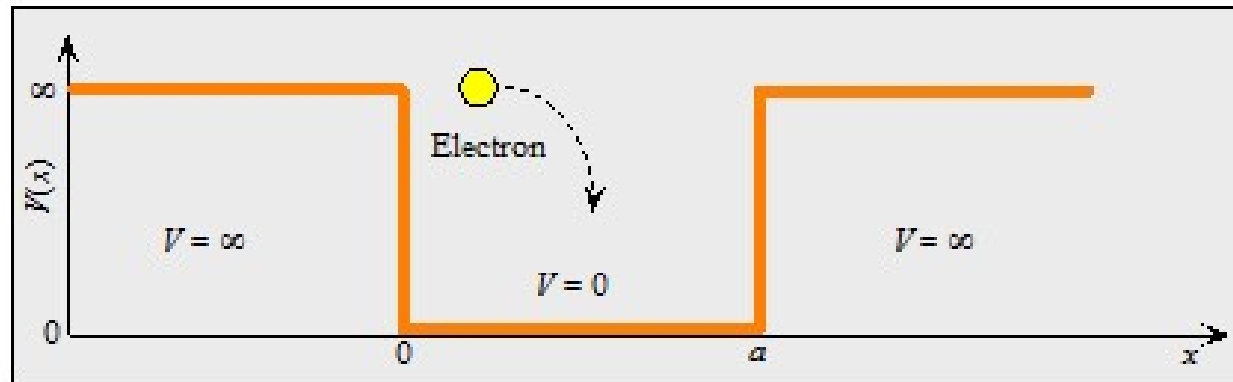
For free electron, $V=0$ $\frac{d^2 \varphi}{dx^2} + \frac{2m}{\hbar^2} E \varphi = \frac{d^2 \varphi}{dx^2} + k^2 \varphi = 0$ where $k^2 = \frac{2m}{\hbar^2} E$

solving this equation, yields $\varphi(x) = A \exp(jkx)$ or $\varphi(x) = B \exp(-jkx)$

- The free electron has a travelling wave solution with a wave number $k = 2\pi/\lambda$
- Energy of the electron $E = KE = \frac{(\hbar k)^2}{2m} = \frac{(p)^2}{2m}$, thus $p = \hbar k = \frac{h}{\lambda}$
- The electron has a well-defined momentum p , uncertainty of $\Delta p = 0$
- The probability distribution for the electron is $|\varphi(x)|^2 = |A \exp(jkx)|^2 = A^2$
- The electron can be anywhere between $-\infty$ and $+\infty$, uncertainty of $\Delta x = \infty$
- Uncertainty principle: can not simultaneous measure the momentum and position



Infinite Potential Well: A Confined Electron



In the well, $V=0$, hence $\frac{d^2\varphi}{dx^2} + \frac{2m}{\hbar^2} E\varphi = \frac{d^2\varphi}{dx^2} + k^2\varphi = 0$ where $k^2 = \frac{2m}{\hbar^2} E$

The solution is $\varphi(x) = A \exp(jkx) + B \exp(-jkx)$

➤ Boundary condition 1: when $x=0$, $\varphi(0)=0$, then we have $A=-B$, therefore $\varphi(x) = 2Aj \sin(kx)$

➤ Boundary condition 2: when $x=a$, $\varphi(a)=0$, then we have $\varphi(a) = 2Aj \sin(ka) = 0$

$$\text{so } ka = n\pi \quad \varphi_n(x) = 2Aj \sin\left(\frac{n\pi}{a}x\right), \text{ where } n=1, 2, 3\ldots \quad E_n = \frac{\hbar^2 k_n^2}{2m}$$

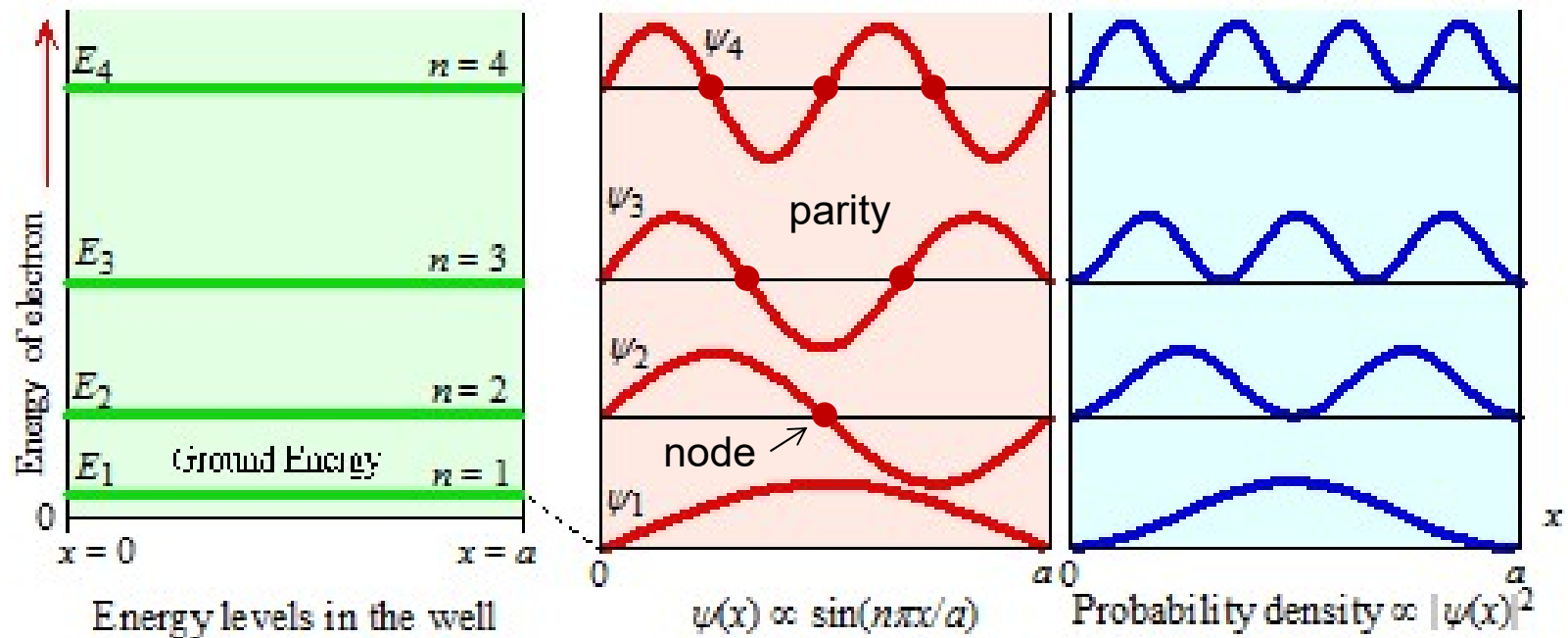
➤ Boundary condition 3: $\int_{x=0}^{x=a} |\varphi(x)|^2 dx = \int_{x=0}^{x=a} \left| 2Aj \sin\left(\frac{n\pi}{a}x\right) \right|^2 dx = 1$, hence $A = \left(\frac{1}{2a}\right)^{1/2}$



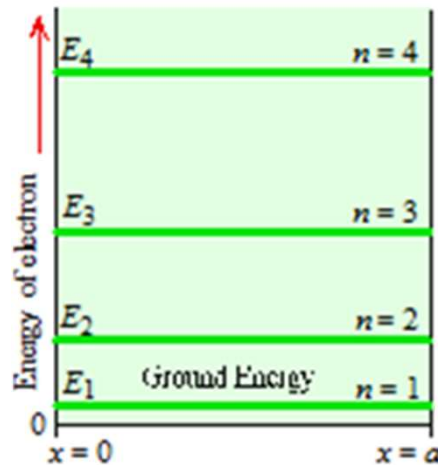
Infinite Potential Well: A Confined Electron

$$\varphi_n(x) = j \left(\frac{2}{a} \right)^{1/2} \sin \left(\frac{n\pi x}{a} \right)$$

$$E_n = \frac{(\hbar k_n)^2}{2m} = \frac{\hbar^2 (n\pi)^2}{2ma^2} = \frac{h^2 n^2}{8ma^2}$$



Quantum Confinement Effect



The smaller the quantum well a , the higher the ground energy and the energy gap are

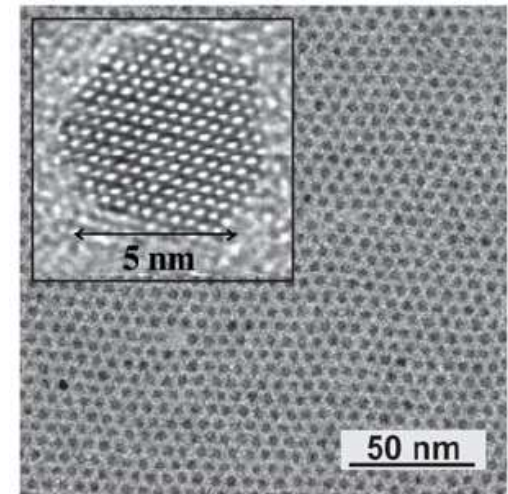
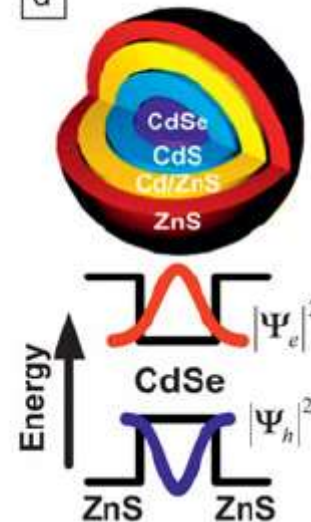
$$E_n = \frac{(\hbar k_n)^2}{2m} = \frac{\hbar^2 (n\pi)^2}{2ma^2} = \frac{h^2 n^2}{8ma^2}$$

a



1.7 nm → 5.0 nm

d



Examples

EXAMPLE 3.7

ELECTRON CONFINED WITHIN ATOMIC DIMENSIONS Consider an electron in an infinite potential well of size 0.1 nm (typical size of an atom). What is the ground energy of the electron? What is the energy required to put the electron at the third energy level? How can this energy be provided?

SOLUTION

The electron is confined in an infinite potential well, so its energy is given by

$$E_n = \frac{h^2 n^2}{8ma^2}$$

We use $n = 1$ for the ground level and $a = 0.1$ nm. Therefore,

$$E_1 = \frac{(6.6 \times 10^{-34} \text{ J s})^2 (1)^2}{8(9.1 \times 10^{-31} \text{ kg})(0.1 \times 10^{-9} \text{ m})^2} = 6.025 \times 10^{-18} \text{ J} \quad \text{or} \quad 37.6 \text{ eV}$$

The frequency of the electron associated with this energy is

$$\omega = \frac{E}{\hbar} = \frac{6.025 \times 10^{-18} \text{ J}}{1.055 \times 10^{-34} \text{ J s}} = 5.71 \times 10^{16} \text{ rad s}^{-1} \quad \text{or} \quad \nu = 9.092 \times 10^{15} \text{ s}^{-1}$$

The third energy level E_3 is

$$E_3 = E_1 n^2 = (37.6 \text{ eV})(3)^2 = 338.4 \text{ eV}$$

The energy required to take the electron from 37.6 eV to 338.4 eV is 300.8 eV. This can be provided by a photon of exactly that energy; no less, and no more. Since the photon energy is $E = h\nu = hc/\lambda$, or

$$\begin{aligned} \lambda &= \frac{hc}{E} = \frac{(6.6 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ m s}^{-1})}{300.8 \text{ eV} \times 1.6 \times 10^{-19} \text{ C}} \\ &= 4.12 \text{ nm} \end{aligned}$$



Heisenberg's Uncertainty Principle

It is impossible to describe with absolute precision events involving individual particles on the atomic scale. Instead, we must speak of the average values of position, momentum, energy of a particle such as an electron.

In any measurement of the position (or energy) and momentum (or time) of a particle, the uncertainties in the two measured quantities are related by

$$\begin{aligned}(\Delta x)(\Delta p) &\geq \hbar \\ (\Delta E)(\Delta t) &\geq \hbar\end{aligned}$$

Quantum Mechanics: Simultaneous measurements of the two quantities are not accurate, and so, we cannot properly speak of the position of an 'e' but must look for the probability of finding an 'e' at a certain point.

i.e., since the traveling wave extends over all space with the same amplitude, the uncertainty Δx in the position of the 'e' is infinite. But, the uncertainty Δp_x in the momentum of the 'e' is zero because λ is well-defined for the traveling wave from the de Broglie relationship $p_x = h/\lambda$

For an 'e' trapped in a 1D infinite PE well, the wave function extends from $x=0$ to a , so the Δx is a . We know the 'e' is within the well, but we cannot pinpoint with certainty exactly where it is. The momentum Δp_x is either $p_x = \hbar k$ or $-\hbar k$, and so, $\Delta p_x = 2\hbar k$. For the ground-state wave function, we have $ka = \pi$, and then $(\Delta x)(\Delta p) = (a)(2\hbar/\pi a) = h$



Examples

EXAMPLE 3.10

THE UNCERTAINTY PRINCIPLE ON THE ATOMIC SCALE Consider an electron confined to a region of size 0.1 nm, which is the typical dimension of an atom. What will be the uncertainty in its momentum and hence its kinetic energy?

SOLUTION

We apply the Heisenberg uncertainty relationship, $\Delta x \Delta p_x \approx \hbar$, or

$$\Delta p_x \approx \frac{\hbar}{\Delta x} = \frac{1.055 \times 10^{-34} \text{ J s}}{0.1 \times 10^{-9} \text{ m}} = 1.055 \times 10^{-24} \text{ kg m s}^{-1}$$

The uncertainty in the velocity is therefore

$$\Delta v = \frac{\Delta p_x}{m_e} = \frac{1.055 \times 10^{-24} \text{ kg m s}^{-1}}{9.1 \times 10^{-31} \text{ kg}} = 1.16 \times 10^6 \text{ m s}^{-1}$$

We can take this uncertainty to represent the order of magnitude of the actual speed. The kinetic energy associated with this momentum is

$$\begin{aligned} KE &= \frac{\Delta p_x^2}{2m_e} = \frac{(1.055 \times 10^{-24} \text{ kg m s}^{-1})^2}{2(9.1 \times 10^{-31} \text{ kg})} \\ &= 6.11 \times 10^{-19} \text{ J} \quad \text{or} \quad 3.82 \text{ eV} \end{aligned}$$



Examples

EXAMPLE 3.11

THE UNCERTAINTY PRINCIPLE WITH MACROSCOPIC OBJECTS Estimate the minimum velocity of an apple of mass 100 g confined to a crate of size 1 m.

SOLUTION

Taking the uncertainty in the position of the apple as 1 m, the apple is somewhere in the crate,

$$\Delta p_x \approx \frac{\hbar}{\Delta x} = \frac{1.05 \times 10^{-34} \text{ J s}}{1 \text{ m}} = 1.05 \times 10^{-34} \text{ kg m s}^{-1}$$

So the minimum uncertainty in the velocity is

$$\Delta v_x = \frac{\Delta p_x}{m} = \frac{1.05 \times 10^{-34} \text{ kg m s}^{-1}}{0.1 \text{ kg}} = 1.05 \times 10^{-33} \text{ m s}^{-1}$$

The quantum nature of the universe implies that the apple in the crate is moving with a velocity on the order of $10^{-33} \text{ m s}^{-1}$. This cannot be measured by any instrument; indeed, it would take the apple $\sim 10^{19}$ years to move an atomic distance of 0.1 nm.



Heisenberg's Matrix Mechanics



The Nobel Prize in Physics 1932
Werner Heisenberg

The Nobel Prize in Physics 1932



Werner Karl
Heisenberg

- Matrix Mechanics utilizing the mathematics of matrices, by Heisenberg
- Wave Mechanics utilizing the wave equations, by Schrödinger

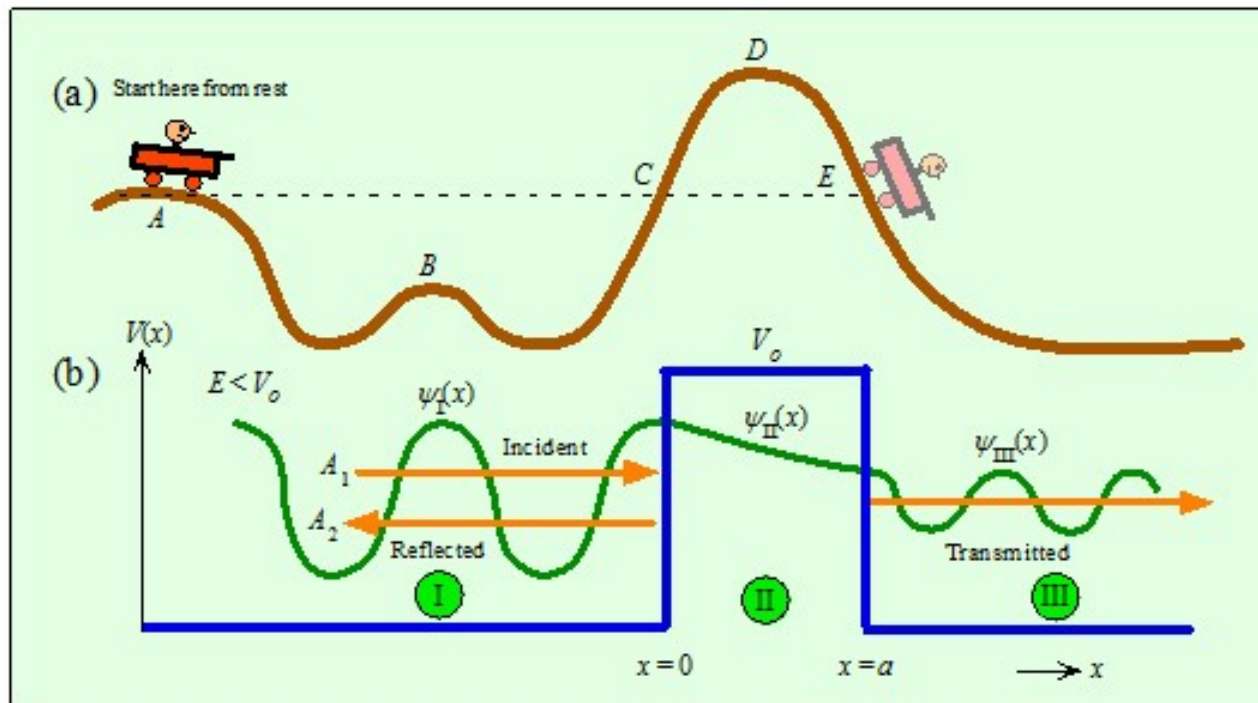
The Nobel Prize in Physics 1932 was awarded to Werner Heisenberg *"for the creation of quantum mechanics, the application of which has, inter alia, led to the discovery of the allotropic forms of hydrogen"*.



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Tunneling Phenomenon: Quantum Leak

In QM, a possibility that the electron will “tunnel” through the potential barrier and appear on the other side (leak through) always exists, unlike in CM.



$$\frac{d^2\phi(x)}{dx^2} + \frac{2m}{\hbar^2}(E - V)\phi(x) = 0$$

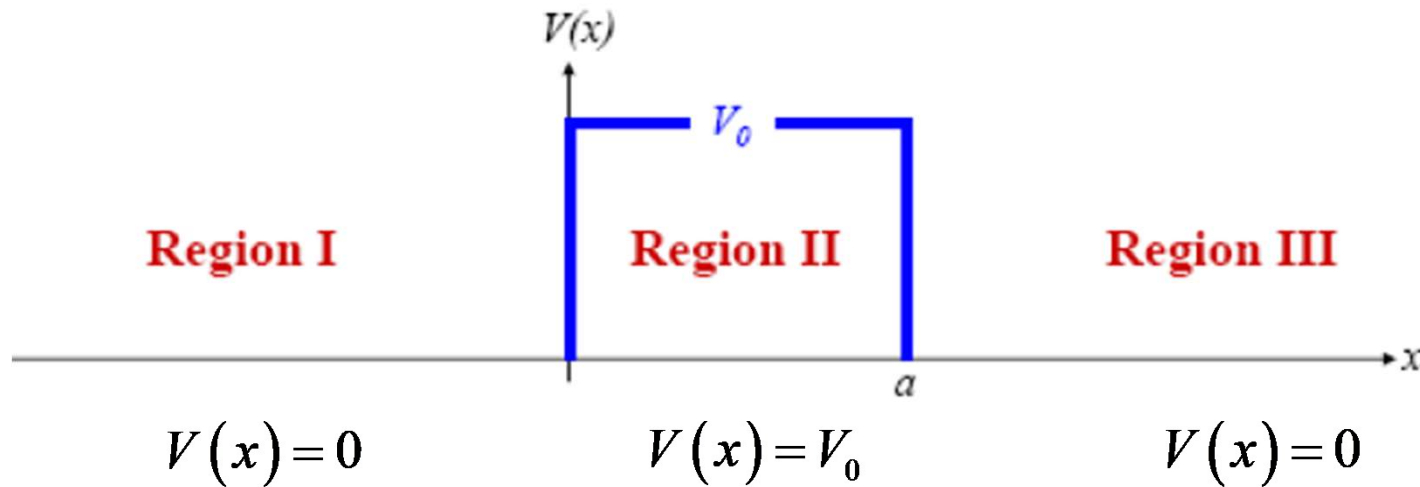
I, III: $E - V = E$

II : $E - V = \text{negative}$



Tunneling Phenomenon: Quantum Leak

Boundary Conditions



$$\psi_I(x) = A_1 \exp(jkx) + A_2 \exp(-jkx)$$

$$\psi_{II}(x) = B_1 \exp(\alpha x) + B_2 \exp(-\alpha x)$$

$$\psi_{III}(x) = C_1 \exp(jkx) + C_2 \exp(-jkx)$$

$$k^2 = \frac{2m}{\hbar^2} E$$

$$\alpha^2 = \frac{2m(V_0 - E)}{\hbar^2}$$



Tunneling Phenomenon: Quantum Leak

The relative probability of the electron from region I through to III, transmission coefficient T

$$T = \frac{|\varphi_{III}(x)|^2}{|\varphi_1(\text{incident})|^2} = \frac{C_1^2}{A_1^2} = \frac{1}{1 + D \sinh^2(a\alpha)} \quad \text{where} \quad \begin{cases} \sinh(x) = \frac{e^x - e^{-x}}{2} \\ D = \frac{V_0^2}{4E(V_0 - E)} \end{cases}$$

For a wide or high barrier, using $a\alpha \geq 1 : \sinh(a\alpha) \approx \exp(a\alpha)/2$

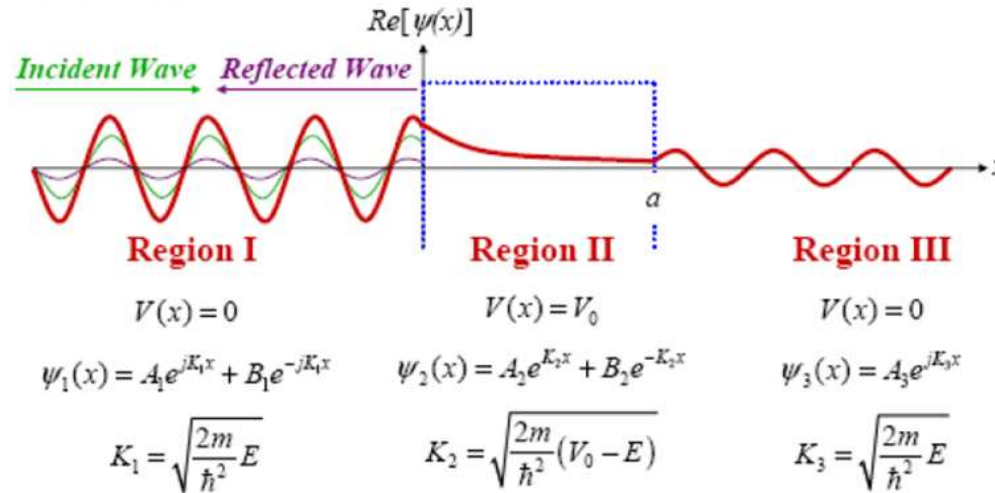
$$T = T_0 \exp(-2a\alpha) \quad \text{where} \quad T_0 = \frac{16E(V_0 - E)}{V_0^2}$$

$$T = 16 \left(\frac{E}{V_0} \right) \left(1 - \frac{E}{V_0} \right) e^{-\sqrt{\frac{8m}{\hbar^2}}(V_0 - E)a}$$

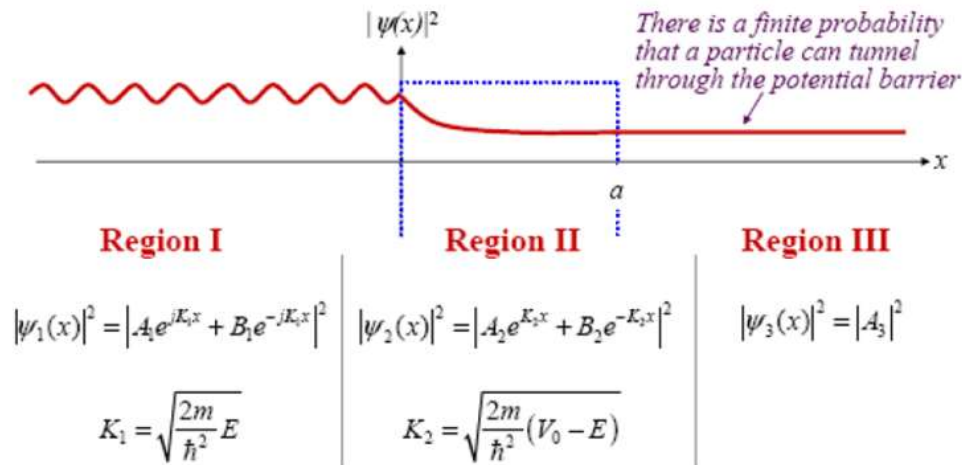


Tunneling Phenomenon: Quantum Leak

Wave functions

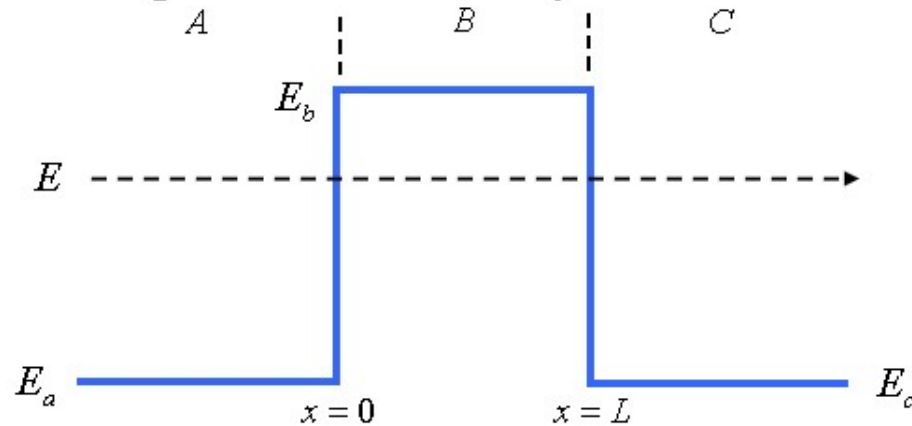


Probability functions



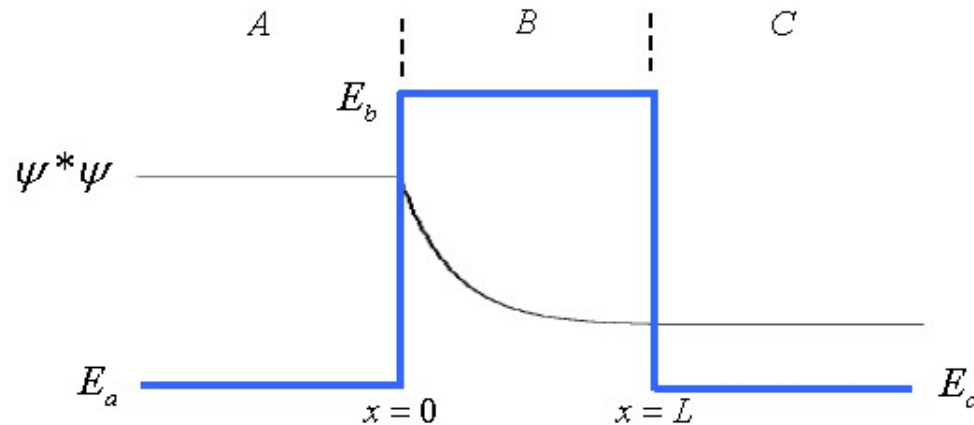
Tunneling Phenomenon: Quantum Leak

Electrons can pass through the energy barrier, even if the region of space is classically forbidden.



An electron approaches a finite potential barrier

B: Classically forbidden region



The probability density function



Tunneling Phenomenon: Quantum Leak

Tunneling probability

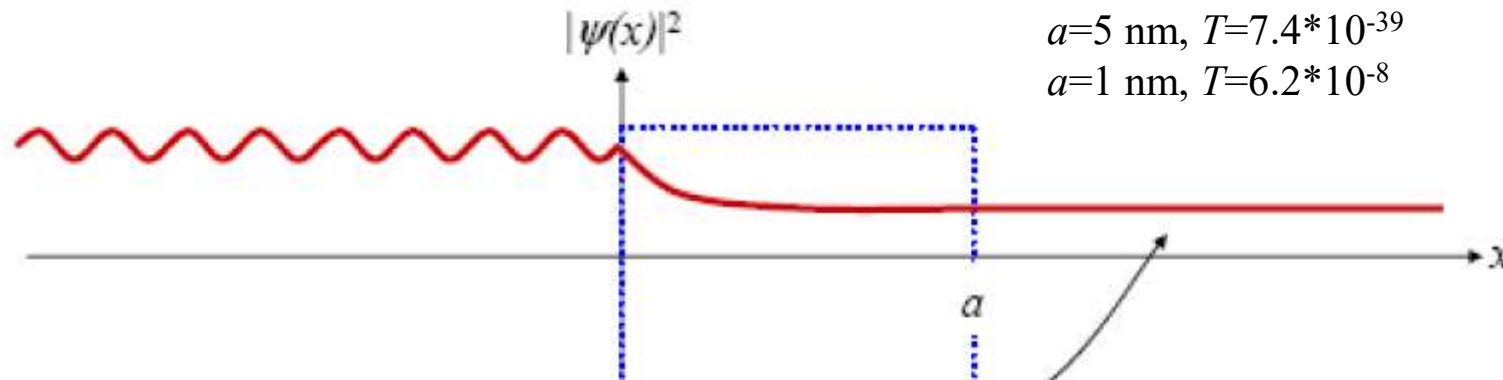
□ Decrease V_0 , T increase

□ Decrease a , T increase

$V_0 = 10 \text{ eV}$, $E = 7 \text{ eV}$

$a = 5 \text{ nm}$, $T = 7.4 \times 10^{-39}$

$a = 1 \text{ nm}$, $T = 6.2 \times 10^{-8}$



The probability that a particle will tunnel through the potential barrier is given by (also referred to as the transmission coefficient or tunneling probability)

$$T \approx 16 \left(\frac{E}{V_0} \right) \left(1 - \frac{E}{V_0} \right) \cdot e^{-\sqrt{\frac{8m}{\hbar^2}}(V_0 - E) \cdot a}$$

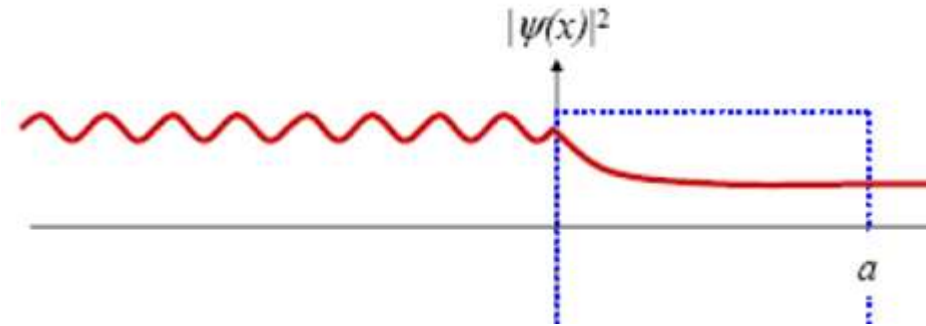
when $V_0 > E$



Can Human Being Tunnel to Past

$$T = 16 \left(\frac{E}{V_0} \right) \left(1 - \frac{E}{V_0} \right) e^{-\sqrt{\frac{8m}{\hbar^2}}(V_0 - E)a}$$

where $V_0 > E$



If $\hbar^2$ is comparable with m
Tunneling can happen



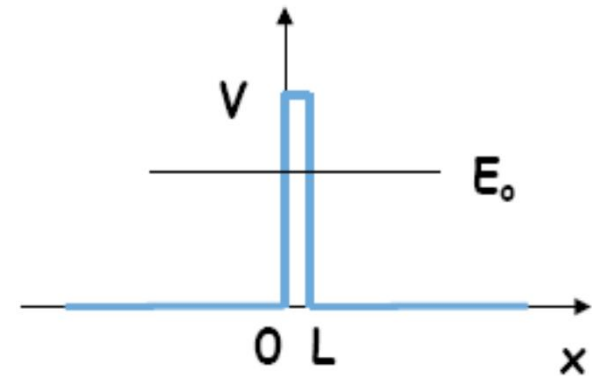
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Multiple Choice Questions

Consider a particle tunneling through a barrier:

1. Which of the following will increase the likelihood of tunneling?

- a. decrease the height of the barrier
- b. decrease the width of the barrier
- c. decrease the mass of the particle



2. What is the energy of the particles that have successfully “escaped”?

- a. < initial energy
- b. = initial energy
- c. > initial energy

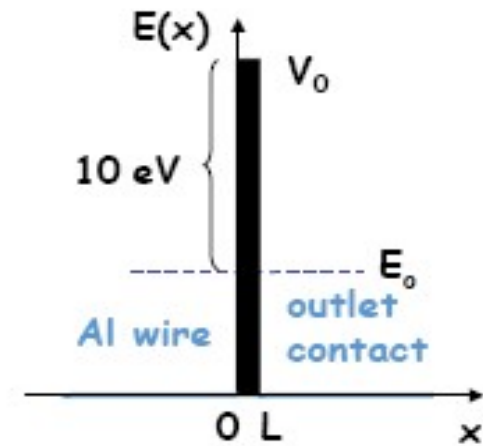
Although the *amplitude* of the wave is smaller after the barrier, no energy is lost in the tunneling process



Example: Al wire contacts

“Everyday” problem:

You’re putting the electrical wiring in your new house, and you’re considering using **Aluminum** wiring, which is cheap and a good conductor. However, you also know that aluminum tends to form an oxide surface layer (Al_2O_3) which can be as much as **several nanometers thick**.



This oxide layer could cause a problem in making electrical contacts with outlets, for example, since it presents a barrier of roughly 10 eV to the flow of electrons in and out of the Al wire.

Your requirement is that your transmission coefficient across any contact must be $T > 10^{-10}$, or else the resistance will be too high for the high currents you’re using, causing a fire risk.

Should you use aluminum wiring or not?

Compute L:

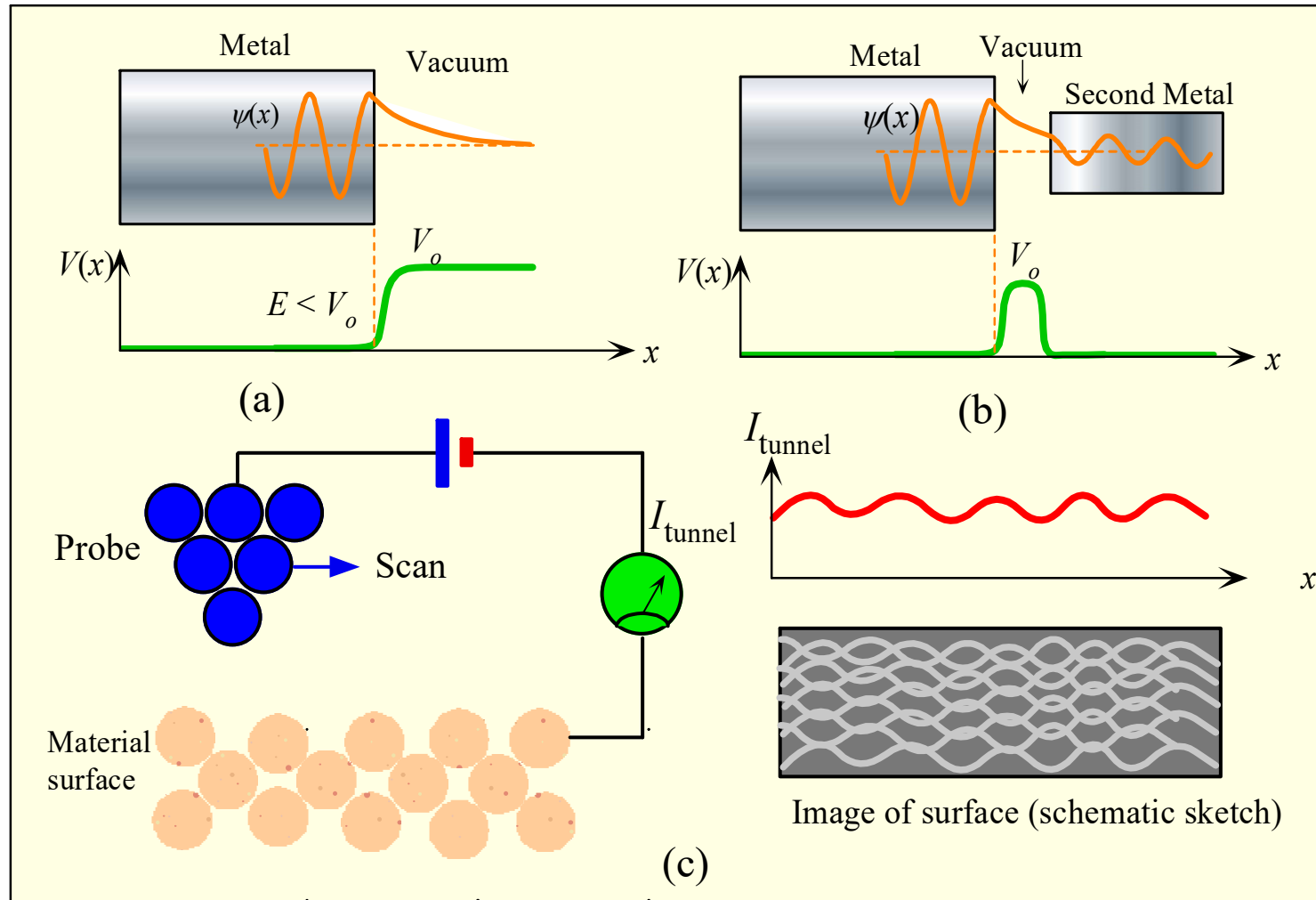
$$T \approx e^{-2\kappa L} \approx 10^{-10} \quad \Rightarrow \quad L \approx -\frac{1}{2\kappa} \ln(10^{-10}) \approx 0.72 \text{ nm}$$

$$\kappa = \sqrt{\frac{2m_e}{\hbar^2}(V_0 - E)} = 2\pi\sqrt{\frac{2m_e}{\hbar^2}(V_0 - E)} = 2\pi\sqrt{\frac{10 \text{ eV}}{1.505 \text{ eV} \cdot \text{nm}^2}} = 16 \text{ nm}^{-1}$$

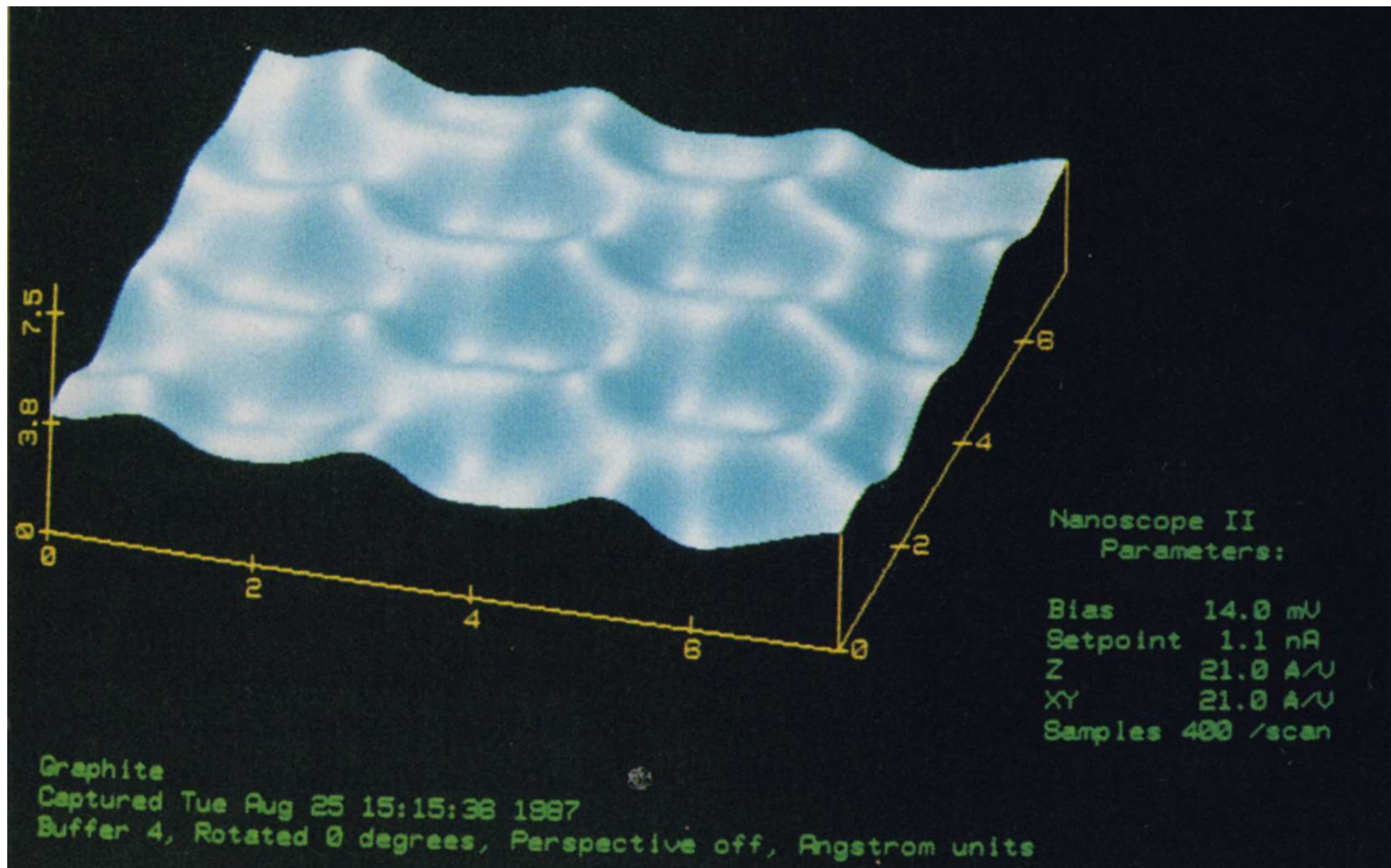
Oxide is thicker than this, so go with Cu wiring! (Al wiring in houses is illegal for this reason)



Tunneling Phenomenon: Quantum Leak



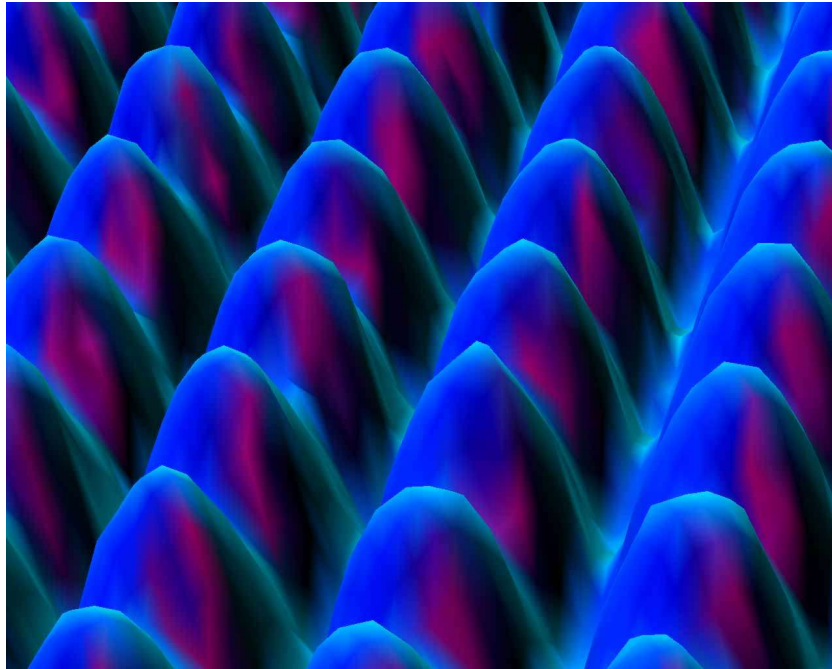
Scanning Tunneling Microscopy (STM)



Scanning Tunneling Microscopy (STM) image of a graphite surface where contours represent electron concentrations within the surface, and carbon rings are clearly visible. Two Angstrom scan.

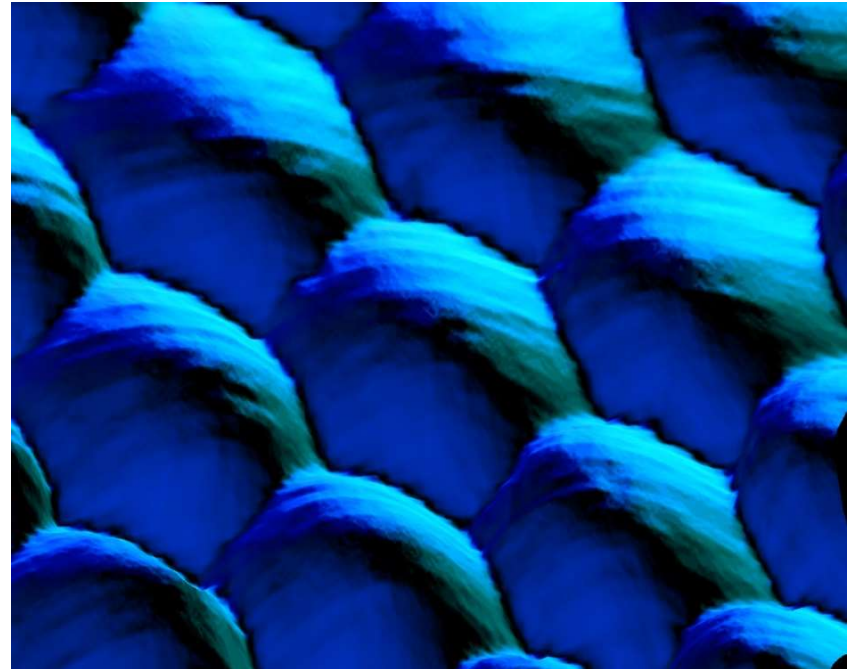


Scanning Tunneling Microscopy (STM)



STM image of Ni (100) surface

SOURCE: Courtesy of IBM

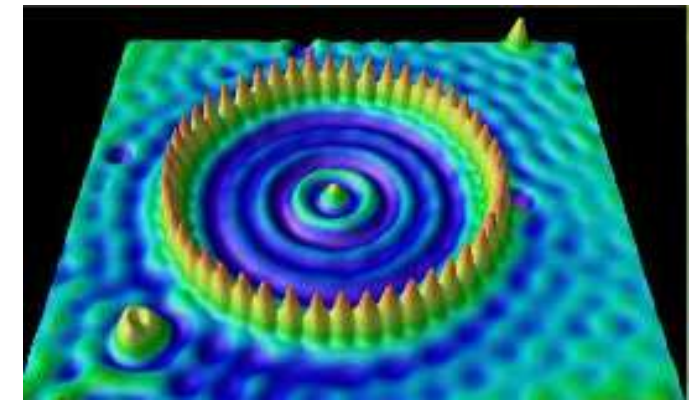
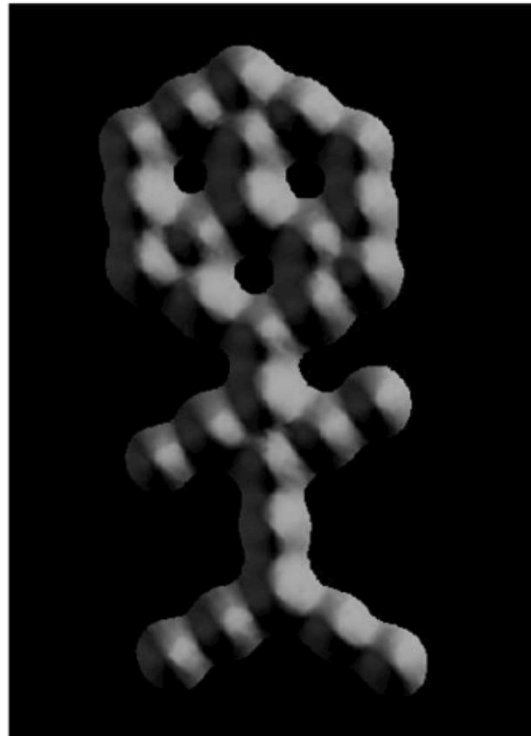
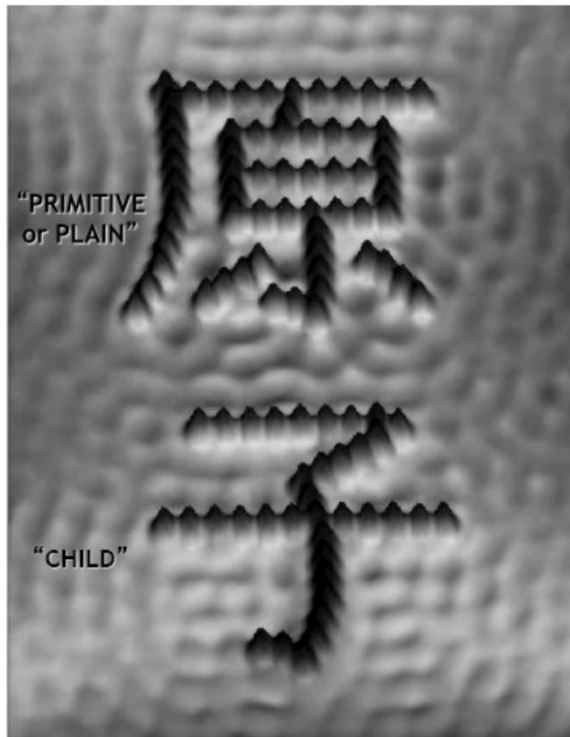
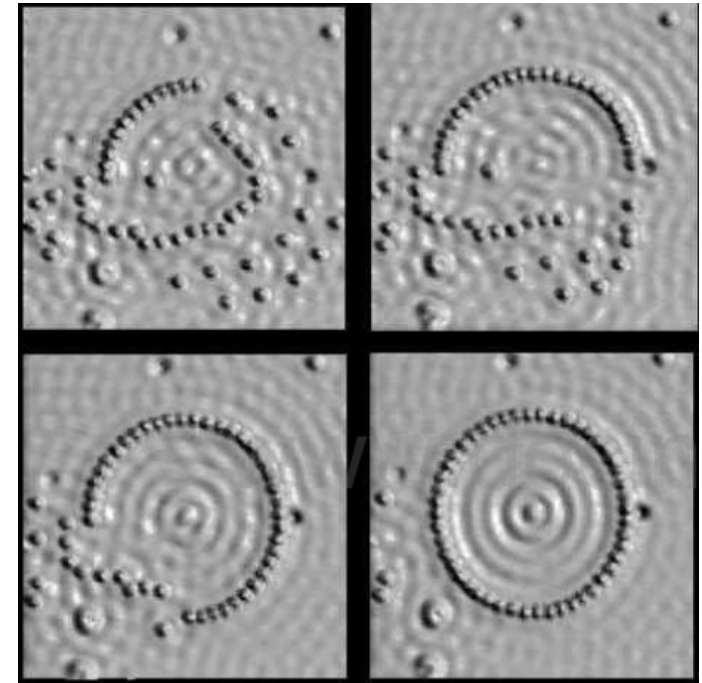
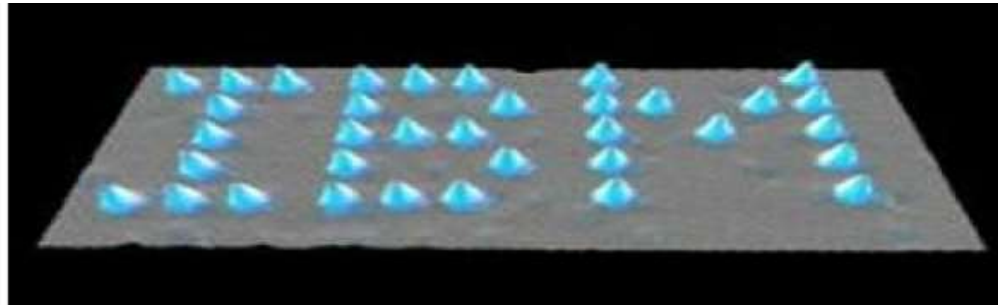


STM image of Pt (111) surface

SOURCE: Courtesy of IBM



Scanning Tunneling Microscopy (STM)



Scanning Tunneling Microscopy (STM)



The Nobel Prize in Physics 1986
Ernst Ruska, Gerd Binnig, Heinrich Rohrer

The Nobel Prize in Physics 1986



Ernst Ruska



Gerd Binnig



Heinrich Rohrer

The Nobel Prize in Physics 1986 was divided, one half awarded to Ernst Ruska *"for his fundamental work in electron optics, and for the design of the first electron microscope"*, the other half jointly to Gerd Binnig and Heinrich Rohrer *"for their design of the scanning tunneling microscope"*.

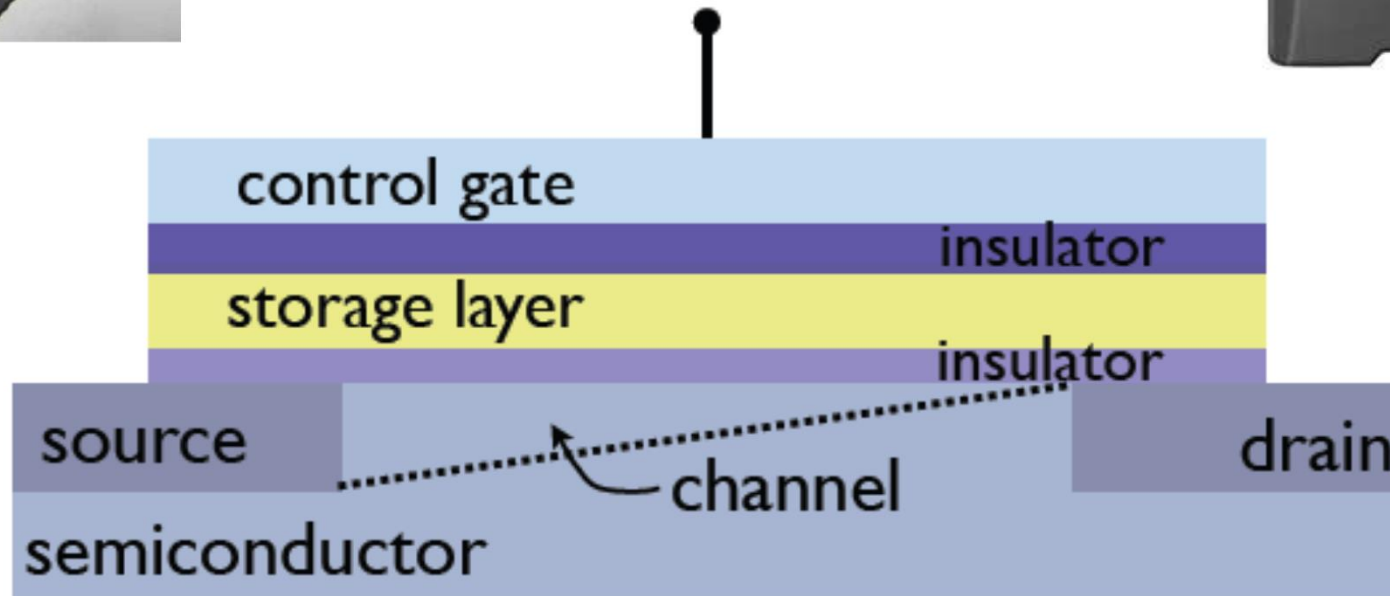


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the basics of Flash



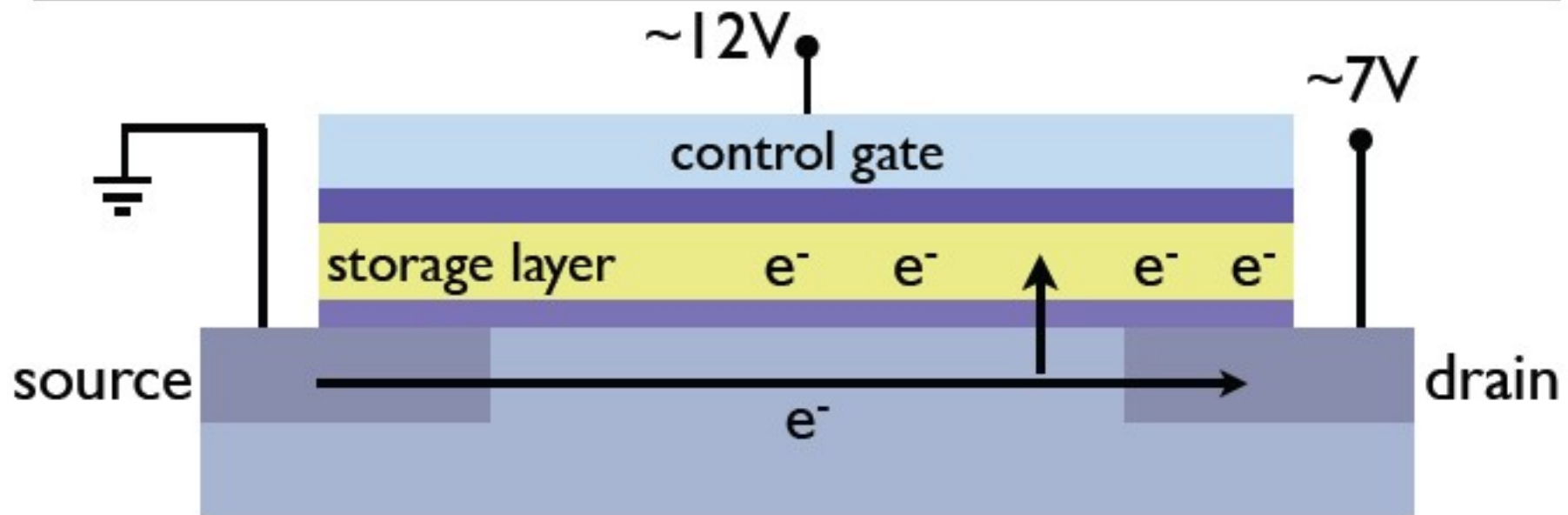
positive V - channel pinched
negative V - channel open



- electrons go between source and drain
- 'control gate' regulates how many
negative voltage stops electrons, positive helps

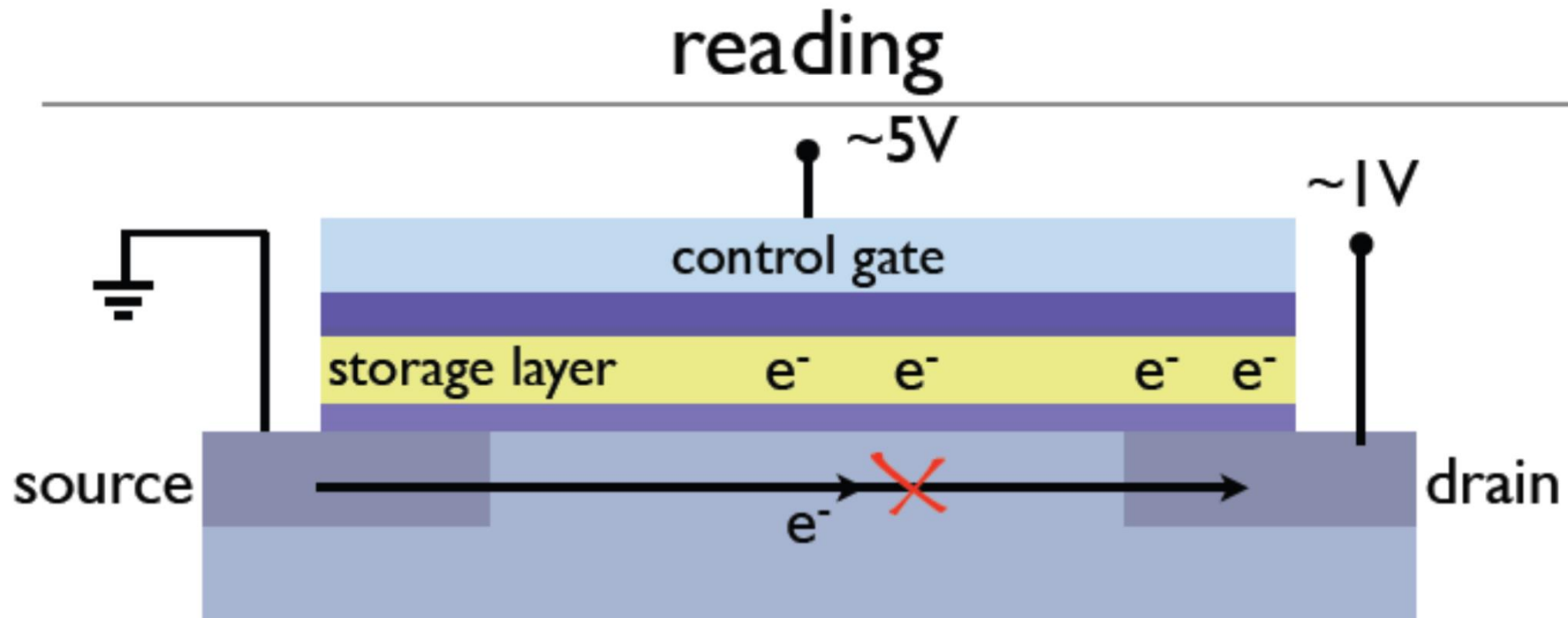


writing



- apply voltage to drain -- pull electrons through channel
- put a positive voltage on control gate
also pulls some electrons into storage layer, via tunneling!
- charge on storage layer = “0” we just wrote information
- electrons do not leak off until a voltage helps them tunnel out!



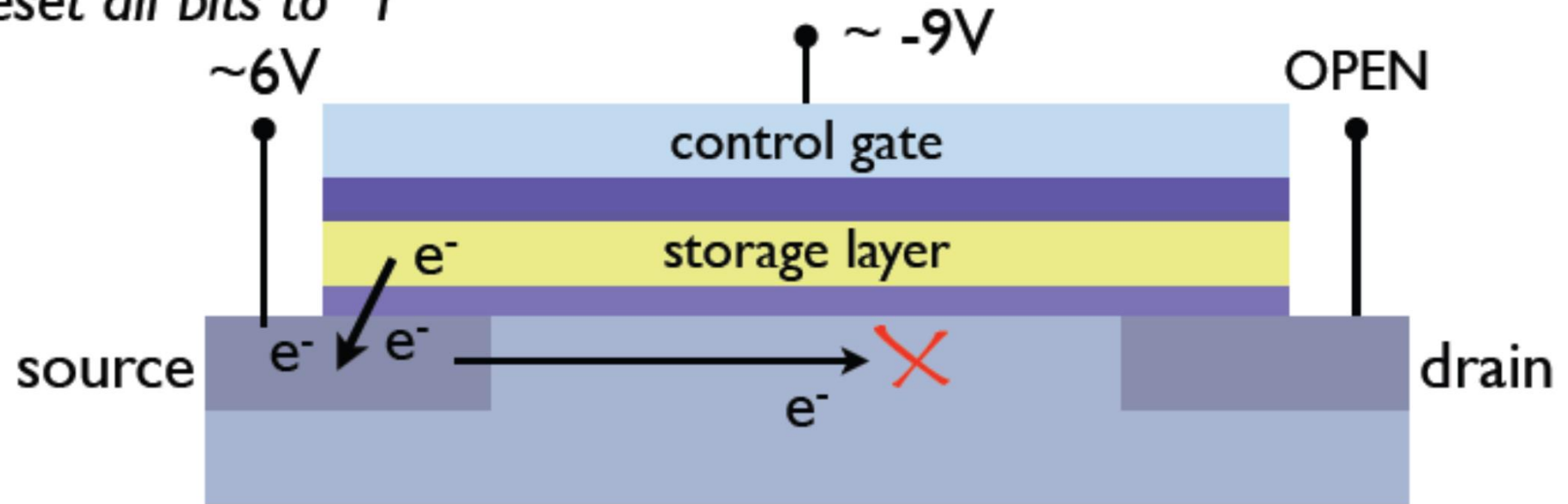


- medium voltage on control, barely open
- if there is charge on the storage layer ...
those electrons repel electrons going from S to D
channel is pinched off = “0”
- no charge on storage layer,
channel is open!
- presence of charge makes current high or low = readout



erasing

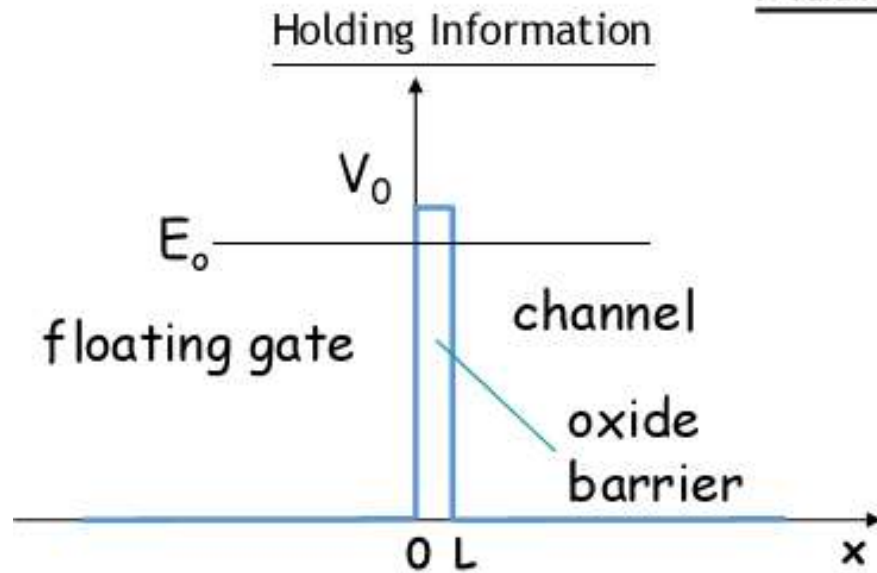
reset all bits to “1”



- negative voltage on control pushes electrons away blocks the channel
- positive voltage on source sucks out the electrons, *only* to the source
- no charge left on storage layer ... just erased our “0” and made “1”



Flash Memory

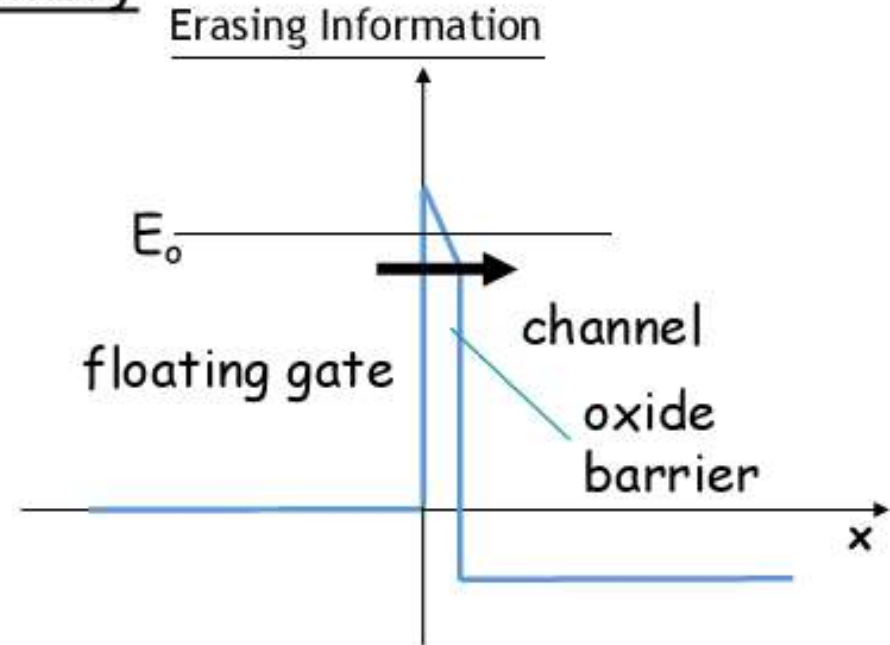


$$T = \left| \frac{F}{A} \right|^2 \approx \frac{16E_o (V - E_o)}{V^2} e^{-2\kappa L}$$

Retention = the ability to hold on to the charge

Tunnel oxide thickness	Time for 20% charge loss
4.5 nm	4.4 minutes
5 nm	1 day
6 nm	½ - 6 years

7-8 nm oxide thickness is the bare minimum, so that the flash memory chip can retain charge in the floating gates for at least 20 years

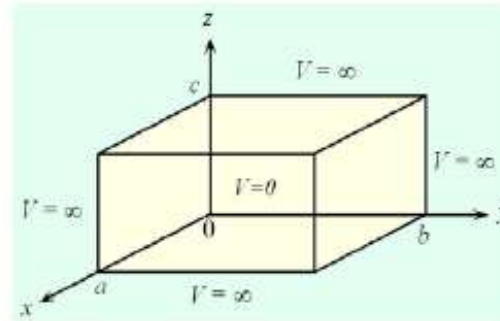


Effective thickness of the tunneling barrier decreases, as the applied voltage bends the potential energy levels



Potential Box: Three Quantum Numbers

To examine the properties of a particle confined to a region of space, we take a 3D space with a volume marked by a, b, c , along the x, y, z axes. The PE is zero inside the space and is infinite on the outside.



What's the behavior of the 'e' in this box ?

Figure 3.19. Electron confined in 3-D by a 3-D infinite PE box

❖ 3-D version of the Schrödinger Eq.

$$\frac{d^2\psi}{dx^2} + \frac{d^2\psi}{dy^2} + \frac{d^2\psi}{dz^2} + \frac{2m}{\hbar^2}(E - V)\psi = 0 \quad (1)$$

- We can solve this by separating the variables via $\psi(x, y, z) = \psi(x)\psi(y)\psi(z)$
and total simple wave-function is simply the product $\psi(x, y, z) = A \sin(k_x x) \sin(k_y y) \sin(k_z z)$
- Applying the boundary conditions at $x=a, y=b, z=c$ to determine the constants k_x, k_y, k_z gives

$$k_x = \frac{n_1\pi}{a} \quad k_y = \frac{n_2\pi}{b} \quad k_z = \frac{n_3\pi}{c} \quad (2)$$



Potential Box: Three Quantum Numbers

- eigenfunctions of the electron, denoted by the quantum numbers n_1, n_2, n_3 , are now given by

$$\psi_{n_1 n_2 n_3}(x, y, z) = A \sin\left(\frac{n_1 \pi x}{a}\right) \sin\left(\frac{n_2 \pi y}{b}\right) \sin\left(\frac{n_3 \pi z}{c}\right)$$

These wavefunctions consist of the products of infinite 1-D PE well-type wave-functions, one for each dimension, and each has its own quantum number n

Each possible eigenfunction can also be labeled a state for the electron

- ❖ We can find the energy of the electron by substituting the wave-function into (1)

$$E = E(k_x, k_y, k_z) = \frac{\hbar^2}{2m} (k_x^2 + k_y^2 + k_z^2)$$

and using (2)
$$E_{n_1 n_2 n_3} = \frac{h^2}{8m} \left(\frac{n_1^2}{a^2} + \frac{n_2^2}{b^2} + \frac{n_3^2}{c^2} \right)$$

- for a square box for $a=b=c=L$

$$E_{n_1 n_2 n_3} = \frac{h^2 (n_1^2 + n_2^2 + n_3^2)}{8mL^2} = \frac{h^2 N^2}{8mL^2}$$

→ In 3D, we have 3 quantum numbers, each one arising from the boundary conditions along one of the coordinates.

